

Literature Review & Elaboration of Problem

PSL Recognizer: Empowering Communication through AI

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1. Abstract

Pakistani Sign Language (PSL) recognition plays a vital role in reducing communication barriers for the deaf community by enabling accessibility in education, workplaces, and everyday interactions. In the initial phase of this project, a smaller PSL dataset was used to develop deep learning-based models for isolated word recognition. Among the proposed architectures, the best-performing model was a custom hybrid based on a **3D Convolutional Neural Network (3DCNN)**, **Residual Blocks**, and a **Bidirectional LSTM**. While the results demonstrated strong initial potential, progress remained constrained by the scarcity of diverse and large-scale PSL video resources.

To strengthen generalization, the dataset was subsequently expanded through new recordings and controlled variations, and the hybrid 3DCNN-based model was retrained on this larger and more diverse data. Despite improved learning on both original and augmented samples, the model continued to exhibit significant overfitting and failed to maintain performance when evaluated on unseen signers. This confirmed that the current approach struggles to learn signer-invariant representations, which are essential for practical, real-world deployment.

Building on these insights, the current research shifts toward developing a **scalable, robust, and signer-independent PSL recognition framework**. Alongside continued refinement of the existing 3DCNN-based approach, extensive investigation is being carried out on **pretrained sign-language and video representation models**. Early exploration indicates that **SignBERT+**, **SignVLM**, and **Gloss-Free Visual Language Pretraining (GF-VLP)** are particularly promising due to their ability to learn rich spatio-temporal and pose-aware features in low-resource settings. The project aims to evaluate and potentially adopt one or two of these pretrained approaches while concurrently addressing the limitations of the existing architecture.

The expected outcome is a **generalized PSL recognition pipeline that performs reliably across unseen signers and diverse environments**, with rapid inference suitable for real-time usage. Rather than pursuing inflated results under controlled conditions, this work prioritizes **robustness, scalability, and long-term extensibility**, forming a strong foundation for future advancements such as sentence-level PSL translation.

2. Background and Justification

2.1 Background of the Existing Work

Pakistan Sign Language (PSL) research has historically suffered from severe data scarcity, lack of standardized benchmarks, and limited real-world validation. The previous FYP group addressed this gap by constructing a PSL dataset consisting of 104 commonly used signs, recorded by three signers and augmented via multiple frame-level transformations. Using this dataset, several deep learning architectures were evaluated, including ViViT, ResNet-18 + LSTM, and a custom 3DCNN + Residual Block + BiLSTM hybrid model. Among these, the hybrid 3DCNN-based architecture achieved the most balanced performance, reaching **83.89% test accuracy** with approximately **9.3 million trainable parameters**. This work established a valuable technical and experimental baseline for PSL word-level recognition.

However, despite achieving strong performance under controlled conditions, a major limitation became evident: the **model generalizes poorly to unseen signers**. When evaluated on videos from new individuals not present during training, recognition accuracy drops sharply. This behavior indicates that the model has learned **signer-specific visual patterns rather than signer-invariant linguistic features**.

2.2 Enhancement and Continuation of the Previous Work (Justification)

The continuation of this research is driven by the need to move beyond the limitations identified in the earlier PSL recognition pipeline, particularly its lack of robustness under signer-independent testing. Although retraining on an expanded dataset improved learning on known data distributions, testing on a limited set of unseen signers revealed that the existing architecture still fails to generalize reliably. This outcome confirms that improved dataset size and hyperparameter optimization alone are insufficient to achieve real-world robustness.

Accordingly, the scope of this work is being expanded in two parallel directions. First, the current 3DCNN-based architecture will continue to be refined to better exploit spatio-temporal feature learning. Second, and more critically, the research now actively explores **pretrained representation-learning approaches**, with a focus on **SignBERT+**, **SignVLM**, and **Gloss-Free Visual Language Pretraining (GF-VLP)**. These frameworks offer fundamentally different, more generalizable learning mechanisms based on pose, motion, and visual–language alignment.

This dual-track strategy is justified by the requirement to achieve **signer-independent performance**, which cannot be addressed through conventional augmentation and retraining alone. By integrating pretrained sign-aware representations with task-specific learning, this research aims to develop a **robust and extensible PSL recognition framework** capable of reliable real-world deployment rather than narrow, dataset-bound performance.

3. Problem Statement

Despite successful development and retraining of a deep learning–based PSL recognition system, the central problem that now defines this research is the **inability of the current architecture to generalize reliably to unseen signers under real-world conditions**. Although the system demonstrates strong learning on curated and augmented datasets, its performance degrades significantly when exposed to signer diversity outside the training distribution. This indicates that the model is primarily learning **signer-specific visual patterns rather than linguistically meaningful, signer-invariant representations** of PSL.

At this stage, the problem is no longer limited to improving isolated accuracy on controlled datasets. Instead, it is the **absence of a robust, scalable, and signer-independent PSL recognition framework** that can function reliably across diverse users, environments, and recording conditions. Without solving this, the system remains impractical for real-world assistive deployment.

3.1 Current Experimental Results and Observed Generalization Behavior

During the initial retraining phase, the expanded PSL dataset was randomly split into **7,856 training samples, 1,447 validation samples, and 621 testing samples**. This split was performed over the full dataset comprising all signers and their corresponding augmentations (96 videos per label: 19 original and 77 augmented). Under this random split, the retrained **3DCNN + Residual Block + BiLSTM** model converged within **25 epochs** and achieved a **test accuracy of 92%**, showing a clear improvement over the earlier performance reported by the previous team.

To evaluate real-world robustness, the trained model was then tested on **60 fully unseen random samples** collected independently from classmates. These samples were not part of training, validation, or augmentation. On this unseen set, the test accuracy **dropped sharply to 20%**, indicating a severe lack of generalization despite strong performance on the randomly split test set.

To further isolate signer bias, a **signer-disjoint dataset split** was constructed from the original dataset, where:

- Data from **4 signers and their augmentations** were used for training,
- **1 signer and their augmentations** was reserved exclusively for validation,
- **1 signer and their augmentations** was reserved exclusively for testing (approximately a **70% / 15% / 15%** split at the signer level).

Under this strictly signer-independent protocol, the model achieved a test accuracy of only **40%**, further confirming that the existing architecture remains highly sensitive to signer identity.

Following discussion with the supervisor, an additional evaluation strategy was tested where the model was trained on the **full dataset**, while validation was performed using

data from **only one excluded signer**, using **5-fold cross-validation over the validation signer**. Under this adjusted protocol, the average validation accuracy increased to approximately **55%**. However, this setup **did not include a fully unseen test set**, and therefore does not represent true signer-independent generalization.

Collectively, these results demonstrate that while the model exhibits strong fitting under random data splits, its performance **degrades significantly under realistic unseen-signer conditions**, providing direct experimental evidence of **overfitting and signer-dependent learning behavior**.

3.2 Limitations of the Current 3DCNN-Based Approach

The retrained 3DCNN + Residual Block + BiLSTM architecture continues to exhibit **strong overfitting**, despite dataset expansion, hyperparameter tuning, and extensive augmentation. Conventional augmentations such as brightness, rotation, noise injection, contrast variation, and geometric transformations improve tolerance to visual distortions but fail to address variations caused by:

- Differences in **hand geometry and finger articulation**,
- **Individual signing styles and motion dynamics**,
- Variations in **body proportions and articulation speed**,
- **Environmental noise, clothing, and background clutter**.

As a result, the current model learns representations that are **dataset-dependent rather than linguistically grounded**, making it unsuitable for signer-independent deployment.

3.3 Exploration and Limitations of Generic Pretrained Video Models

As part of this semester’s investigation, the project initially explored **generic pretrained action-recognition backbones**, including **I3D, X3D, and SlowFast**, which are trained on large-scale datasets such as Kinetics. Although these models provide strong **general spatio-temporal representations**, they are optimized for **full-body human actions** such as running, jumping, and sports activities.

Through theoretical analysis and early architectural evaluation, it became evident that these models are **not well-aligned with the core requirements of sign language recognition**, for the following reasons:

- They emphasize **global body motion** rather than fine-grained **hand and finger articulation**, which is the primary semantic carrier in PSL.
- Their spatial resolution at deeper layers is often too coarse for **finger-level discrimination**.
- They rely heavily on **appearance- and context-based cues**, which increases susceptibility to signer identity, clothing, and background biases.
- When combined with hand keypoint extractors such as **MediaPipe**, the pipeline becomes unstable in real-world scenarios due to:
 - Occlusions from loose clothing,
 - Motion blur during fast signing,

- Background clutter,
- Varying camera angles and lighting,
which significantly degrade keypoint accuracy and propagate noise into the recognition system.

These limitations indicate that **generic action-recognition pretraining alone is insufficient** for building a signer-independent PSL recognition system.

3.4 Shift Toward Sign-Focused Pretrained Representation Learning

As a result of these limitations, the research problem has now evolved toward identifying and leveraging **sign-language-specific pretrained representation frameworks** that are explicitly designed to model:

- **Hand articulation and pose dynamics,**
- **Temporal linguistic structure,**
- **Signer-invariant motion patterns,**
- **Low-resource learning scenarios.**

Based on early-stage literature review and architectural evaluation, **SignBERT+**, **SignVLM**, and **Gloss-Free Visual Language Pretraining (GF-VLP)** have emerged as the most promising candidates due to their:

- Pose-aware and hand-focused modeling strategies (SignBERT+),
- CLIP-based visual-temporal alignment for low-data learning (SignVLM),
- Direct video-to-language learning without gloss dependency (GF-VLP).

These models introduce fundamentally different learning mechanisms compared to RGB-only CNN-LSTM pipelines and offer **strong theoretical grounding for signer-independent generalization**.

3.5 Revised Research Problem

After completing the foundational implementation and retraining phase, the **revised central problem of this research** is:

To design and develop a signer-independent, scalable, and real-time-capable PSL recognition architecture by integrating sign-focused pretrained representations with task-specific learning, while also improving the robustness of the existing 3DCNN-based framework.

This problem explicitly targets real-world deployment constraints rather than controlled laboratory performance.

3.6 Key Technical and Research Challenges

The major challenges that arise during this research include:

- Learning **signer-invariant representations** from a **limited PSL dataset**,
- Overcoming failures of **conventional hand keypoint extraction** in noisy environments,
- Selecting and **adapting large pretrained sign models** under limited computational resources,
- Balancing **model complexity with real-time inference requirements**,
- Designing an architecture that remains **extendable toward sentence-level PSL translation**,
- Preventing overfitting while maintaining **high discriminative power**.

4. Literature Review

4.1 Signer-Independent Sign Language Recognition and Generalization Challenges

A comprehensive study by Marc, Dane, and colleagues on signer-independent sign language recognition using the LSA64 dataset demonstrates the severity of domain shift when models are evaluated on unseen signers. The authors report that architectures trained under signer-dependent conditions achieve high accuracy, yet their performance collapses when tested on signers excluded from training. Their analysis shows that visual appearance, signer-specific motion patterns, and background biases cause models to learn identity-related shortcuts instead of linguistically meaningful motion primitives. The study establishes that signer-independent evaluation is essential for real-world deployability and that conventional CNN–LSTM pipelines are highly vulnerable to overfitting under limited signer diversity [1]. These findings directly align with the generalization failures observed in our PSL experiments.

4.2 Signer-Diversity–Driven Data Augmentation for Robust Recognition

Recent work by Honghao Fu, Liang Zhang, and their team on signer-diversity–driven data augmentation proposes targeted augmentation strategies that explicitly disrupt signer identity cues rather than only visual appearance. Unlike traditional brightness, rotation, and noise-based augmentation, this approach introduces identity-level diversity through adversarial and style-based transformations that better simulate unseen signers. Their results show significant improvements in signer-independent performance across multiple benchmarks. This work confirms that appearance-only augmentations are insufficient for solving signer generalization and that signer-aware augmentation is a necessary step toward robust sign recognition and translation [2]. However, even signer-aware augmentation alone does not fully address the limitations of representation learning in low-resource sign language settings.

4.3 Generic Pretrained Video Action Models and Their Limitations for Sign Language

Following the limitations observed with augmentation-based strategies, many studies have explored **generic pretrained video action-recognition models** such as **I3D**, **X3D**, and **SlowFast**, which are trained on large-scale datasets like **Kinetics**. I3D inflates 2D CNN filters into 3D to jointly model spatial and temporal features [7], X3D proposes efficient depth-wise scalable 3D architectures for real-time video understanding [8], while SlowFast introduces dual-pathway temporal modeling to capture both slow semantic motion and fast dynamic movements [9]. Although these models provide strong general spatio-temporal representations for full-body human actions, they are fundamentally optimized for activities such as sports, gestures, and locomotion. Their reliance on **global body motion, coarse spatial resolution at deeper layers, and strong contextual cues** makes them poorly aligned with the requirements of fine-grained **hand- and finger-centric sign language recognition**. Moreover, when paired with hand keypoint extractors such as MediaPipe, these models suffer from instability in the presence of occlusion, motion blur, background clutter, and clothing variations. This limits their ability to learn robust signer-invariant features, motivating the need for **sign-specific pretrained architectures**.

4.4 SignBERT: Hand-Model-Aware Self-Supervised Pretraining for Isolated SLR

The SignBERT framework introduces the first large-scale **hand-model-aware self-supervised pretraining approach** for isolated sign language recognition. It treats hand poses as visual tokens and embeds 2D joints extracted via MMPose with gesture-state representations, temporal positional encoding, and left-right chirality tokens. A transformer encoder models intra-hand and inter-hand dependencies, while a MANO-based decoder reconstructs physically plausible 3D hand meshes under masked reconstruction supervision. The model is pretrained on large unlabeled collections from NMFs-CSL, SLR500, MSASL, and WLASL using masked joint, masked frame, and identity modeling strategies. Fine-tuning replaces the reconstruction head with a classification head, resulting in strong performance across multiple isolated SLR benchmarks. This work establishes pose-aware representation learning as a powerful mechanism for achieving signer-invariant recognition [3].

4.5 SignBERT+: Extending Pose-Based Pretraining to Full Sign Language Understanding

SignBERT+ extends the original SignBERT framework beyond isolated recognition to full **sign language understanding**, including continuous recognition and sign-to-text translation. It introduces **clip-level temporal masking** to enforce long-range temporal reasoning and incorporates arm joints for improved global spatial context. Task-specific heads are added, including CTC-based decoders for continuous SLR and autoregressive transformers for SLT. Pretraining is conducted on over 230,000 unlabeled samples with enhanced regularization on MANO pose and shape parameters. Evaluations on MSASL, WLASL, SLR500, and RWTH-Phoenix demonstrate consistent improvements over SignBERT, confirming the importance of hierarchical, pose-centric pretraining for signer-independent robustness [4].

4.6 SignVLM: CLIP-Based Visual–Temporal Transformer for Low-Data Sign Recognition

Luqman (2025) introduces **SignVLM**, a visual–temporal sign recognition framework that integrates a pretrained **CLIP visual encoder** with a transformer-based temporal decoder. Each video frame is encoded using CLIP, which has been pretrained on hundreds of millions of image–text pairs, enabling strong spatial understanding without task-specific supervision. The temporal decoder models motion using temporal convolutions, cross-frame attention, and multi-head self-attention. This architecture significantly reduces dependence on large labeled sign datasets and is particularly suited to low-resource environments such as PSL. Benchmark analysis shows that pose-based and enhanced temporal models consistently outperform pure RGB pipelines across ASL, Arabic, LSA64, and AUTSL datasets, reinforcing the relevance of CLIP-based temporal modeling for scalable sign recognition [5].

4.7 Gloss-Free Visual-Language Pretraining for End-to-End Sign Translation

The Gloss-Free Sign Language Translation with Visual-Language Pretraining (GFSLT-VLP) framework removes the dependency on gloss annotations by introducing a CLIP-inspired **visual–language contrastive pretraining strategy** for direct video-to-text translation. A two-stage architecture is used: Stage 1 aligns visual and textual embeddings using contrastive learning and masked reconstruction, while Stage 2 converts the pretrained components into a full encoder–decoder translation system. Trained with strong geometric, photometric, and temporal augmentations, the framework achieves significant improvements over previous gloss-free methods on PHOENIX14T and CSL-Daily and even rivals gloss-based systems. This work shows that visual-language pretraining provides strong semantic grounding for sign videos and supports future extension of PSL recognition toward sentence-level translation [6].

5. Appendices

Appendix A: PSL Dataset Composition

This appendix contains details related to dataset construction, including:

- Number of PSL classes (104),
- Signer distribution across training and testing,
- Data augmentation strategies applied,
- Frame sampling and preprocessing details.

Appendix B: Model Training Configuration

Includes:

- Network hyperparameters (learning rate, dropout, batch size),
- Number of epochs for baseline and retraining phases,
- GPU/CPU resources utilized during training,
- Evaluation metrics and loss functions.

Appendix C: Performance Analysis

Includes:

- Confusion matrices for baseline and retrained models,
- Per-class accuracy and F1 scores,
- Cross-signer testing breakdown.

6. References

- [1] M. Marais, “Investigating Signer-Independent Sign Language Recognition on the LSA64 Dataset” *Proc. Southern Africa Telecommunication Networks and Applications Conference (SATNAC)*, 2022.
- [2] Honghao Fu, “Signer Diversity-driven Data Augmentation for Signer-Independent Sign Language Translation,” Xiamen University, 2024.
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- [4] Weichao Zhao, “SignBERT+: Hand-model-aware Self-supervised Pre-training for Sign Language Understanding,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2023.
- [5] Luqman, “SignVLM: CLIP-Based Visual–Temporal Transformer for Low-Data Sign Recognition,” 2025.
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- [7] Joao Carreira, “Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset,” *Cornell University*, 2018.
- [8] Christoph Feichtenhofer, “SlowFast Networks for Video Recognition,” *Proc. IEEE Int. Conf. on Computer Vision (ICCV)*, 2019.
- [9] Christoph Feichtenhofer, “X3D: Expanding Architectures for Efficient Video Recognition,” *Facebook AI Research (FAIR)*, 2020.

7. Bibliography

The bibliography includes foundational and supplementary material consulted during the course

of this research, including:

- Books on deep learning and computer vision,
- Research articles on video classification and transformer architectures,
- Technical documentation on MediaPipe, OpenPose, and MMPose,
- Online repositories for SignBERT, SignVLM, and GFSLT-VLP,
- Surveys on sign language recognition and translation.